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PROJECT REPORT

Design and Implementation of a Web Server on Raspberry Pi

Description

This project demonstrates the development of a web server using the Raspberry Pi. The Raspberry Pi acts as a mini computer that hosts a web application, allowing users to access and control connected devices through a browser over a local network. It shows how embedded systems can be used to create low-cost and efficient web-based control systems.

Components Used

Raspberry Pi 4

Micro SD Card (16GB or higher)

Power Adapter

Ethernet Cable / WiFi Connection

Program

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Alex Morgan Portfolio</title>

<!-- Google Font -->
```

```
<link
href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600;700&display=swap"
rel="stylesheet">


<style>
*{
   margin:0;
   padding:0;
   box-sizing:border-box;
   font-family: 'Poppins', sans-serif;
}

body{
   background:#f5f5f5;
   color:#333;
}

/* Navbar */
header{
   background:#0b1c2d;
   padding:20px 10%;
   display:flex;
   justify-content:space-between;
   align-items:center;
}

.logo{
   color:#fff;
   font-size:28px;
   font-weight:600;
}
```

```
nav a{
   color:#fff;
   text-decoration:none;
   margin-left:30px;
   font-weight:400;
   transition:0.3s;
}

nav a:hover{
   color:#4f8cff;
}

/* Hero Section */
.hero{
   height:90vh;
   background: linear-gradient(135deg,#4f8cff,#6a5cff);
   display:flex;
   flex-direction:column;
   justify-content:center;
   align-items:center;
   text-align:center;
   color:#fff;
   padding:0 20px;
}

.hero h1{
   font-size:55px;
   font-weight:700;
   margin-bottom:20px;
}
```

```
.hero p{
   font-size:20px;
   margin-bottom:30px;
}

.btn{
   padding:15px 35px;
   background:#111;
   color:#fff;
   border:none;
   border-radius:40px;
   font-size:16px;
   cursor:pointer;
   transition:0.3s;
}

.btn:hover{
   background:#333;
}

/* Section General */
section{
   padding:80px 10%;
   text-align:center;
}

section h2{
   font-size:36px;
   margin-bottom:30px;
}
```

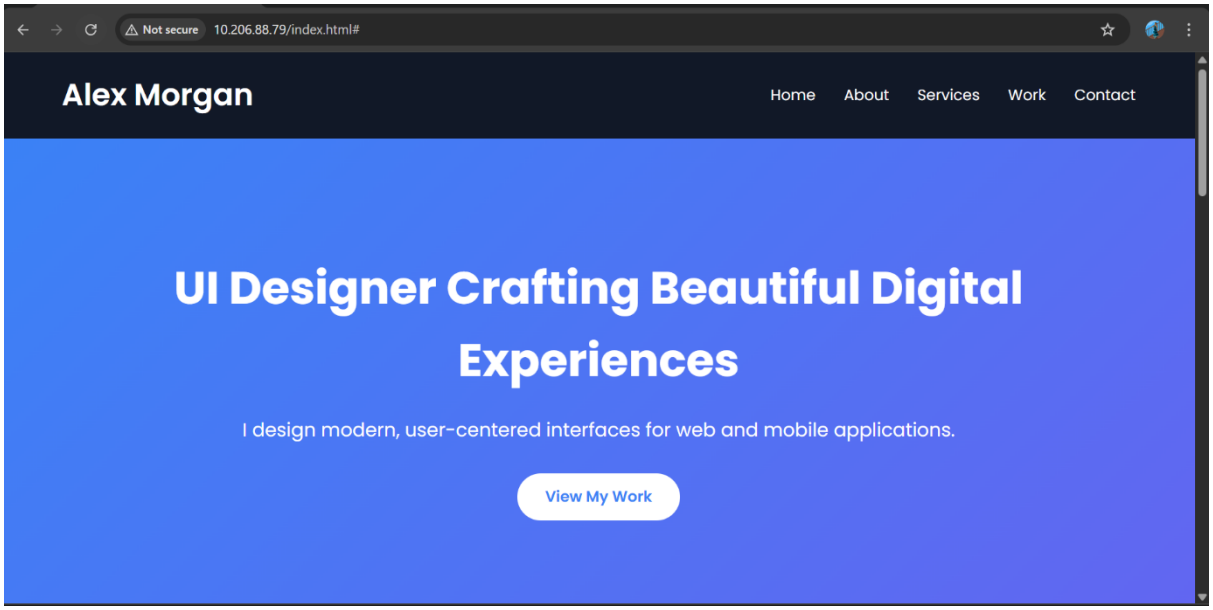
```
/* About */
.about p{
   max-width:700px;
   margin:auto;
   font-size:18px;
   line-height:1.8;
}

/* Services */
.services-container{
   display:flex;
   justify-content:space-between;
   gap:20px;
   margin-top:40px;
   flex-wrap:wrap;
}

.service-box{
   background:#fff;
   padding:40px 20px;
   border-radius:15px;
   flex:1;
   min-width:280px;
   box-shadow:0 5px 15px rgba(0,0,0,0.05);
   transition:0.3s;
}

.service-box:hover{
   transform:translateY(-10px);
}
```

```
.service-box h3{
   margin-bottom:15px;
}

/* Portfolio */
.portfolio{
   background:#f0f0f0;
}

/* Responsive */
@media(max-width:768px){
   .hero h1{
      font-size:35px;
   }
   nav{
      display:none;
   }
}
</style>
</head>

<body>

<header>
   <div class="logo">Alex Morgan</div>
   <nav>
      <a href="#">Home</a>
      <a href="#about">About</a>
      <a href="#services">Services</a>
      <a href="#portfolio">Work</a>
      <a href="#">Contact</a>
```

```
    </nav>
  </header>

  <section class="hero">
    <h1>UI Designer Crafting Beautiful Digital Experiences</h1>
    <p>I design modern, user-centered interfaces for web and mobile applications.</p>
    <button class="btn">View My Work</button>
  </section>

  <section class="about" id="about">
    <h2>About Me</h2>
    <p>
      I’m a passionate UI Designer with 5+ years of experience creating visually appealing
      and user-friendly digital products. My design philosophy focuses on clarity,
      simplicity, and usability.
    </p>
  </section>

  <section class="services" id="services">
    <h2>Services</h2>
    <div class="services-container">
      <div class="service-box">
        <h3>Web UI Design</h3>
        <p>Clean, modern interfaces designed for usability and engagement.</p>
      </div>
      <div class="service-box">
        <h3>Mobile App Design</h3>
        <p>Responsive and intuitive app designs for iOS and Android.</p>
      </div>
      <div class="service-box">
        <h3>Prototyping</h3>
```

```
        <p>Interactive wireframes and prototypes for seamless collaboration.</p>
      </div>
    </div>
  </section>

  <section class="portfolio" id="portfolio">
    <h2>Portfolio</h2>
  </section>

  </body>
  </html>
```

Output



Result

The web server was successfully configured on the Raspberry Pi. The web page was accessible through a browser using the Raspberry Pi's IP address. This project demonstrates that Raspberry Pi can function as a compact, cost-effective web server for embedded and IoT applications.